Hybrid Adaptive Transformer Differential Protection

An Intelligent DWT–LSTM Framework

Final Thesis Defense — B.Sc. in EEE

Date: 7th June, 2026

1. **Md Moinul Haque** *ID: **202216206***
2. **Md Reyaz Uddin** *ID: **202216211***

Cdre A N M Didarul Alam, (L), NUP, psc, BN

Dean, Faculty of Maritime Business Studies,
Bangladesh Maritime University

Department of Electrical, Electronic and Communication Engineering (EECE)
Military Institute of Science and Technology (MIST)

Outline

- 1. Research Motivation
- 2. Comparative & Associative Literature
- 3. Research Objectives
- 4. Proposed Method / Model Framework
- 5. Implementation (Simulink + ONNX deployment)
- 6. Result Analysis
- 7. Completion of Research Aims
- 8. Justification of Course Outcome (CO) Achievement
- 9. Conclusion
- 10. Future Prospects

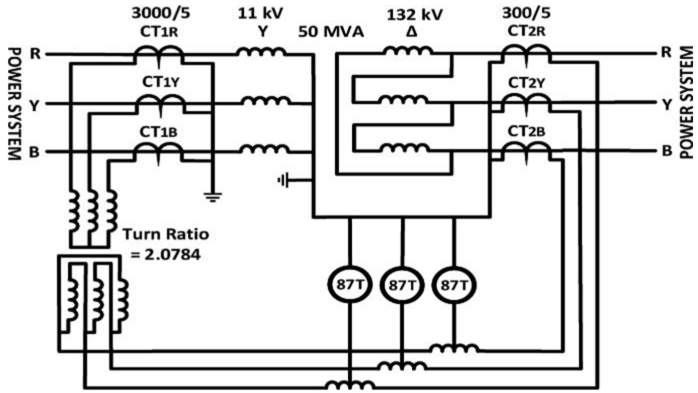
Overview of Differential Protection using 87T Relay

The Basic Principle

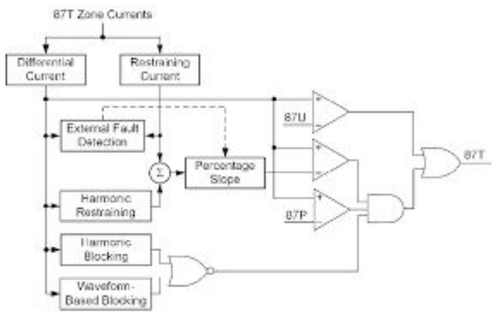
- Kirchhoff's Current Law: current flowing INTO the transformer should equal current OUT.
- The relay measures both sides using Current Transformers (CTs).
- It computes: I_diff = I_HV – I_LV (after correcting for winding ratio).
- If I_diff exceeds a pickup threshold → the relay sends a TRIP command.

The dual-slope characteristic

- A restraint current I_rest scales the threshold automatically during high through-fault current.
- Slope-1 region: low through-current → sensitive pickup.
- Slope-2 region: high through-current → raised threshold to avoid false trips.



(a)



(b)

(c)

Fig: (a) Differential Protection Circuit (b) Logic Diagram (c) Packaging of Siemens SIPROTEC 7UT87 modular transformer differential protection relay

Research Motivation

• Why this matters

- Power transformers are the costliest substation asset (USD 1–3 M per 132/11 kV unit).
- Differential protection (ANSI 87T) is the primary protection element — Kirchhoff's current law inside the protected zone.
- Trip command must be issued in less than one power-system cycle (< 20 ms at 50 Hz).
- Mis-operation impact: prolonged outage, oil-fire risk, cascading grid failure.

• Where the conventional 87T fails

- Magnetizing inrush — 6–10× rated current pulled on one side ⇒ appears as differential ⇒ false trip.
- Modern grain-oriented cores: 2nd-harmonic content $H_2 < 15\ \%$ ⇒ classical restraint criterion fails.
- CT saturation on external faults → clipped secondary ⇒ spurious I_diff.
- Slope-2 bias and fixed thresholds cannot resolve both vulnerabilities at once.
- ***⇒ Need an intelligent, data-driven supervisory layer.***

Comparative & Associative Literature

- **Evolution of the field**

- 1960s — fixed harmonic restraint (2nd / 5th)
- 1990s — adaptive pickup & dual-slope bias
- 2000s — wavelet-based time-frequency features
- 2010s — wavelet + shallow ANN / SVM
- 2020s — end-to-end deep learning (CNN, LSTM, DNN)
- **This work — Hybrid DWT–LSTM with adaptive 87T**

- **Representative recent work**

- Afrasiabi et al. (2020, IEEE TII) — accelerated DNN; limited feature set.
- Key et al. (2025, IEEE Access) — dual-DNN time+freq fusion; no temporal context.
- Alhamd et al. (2024) — improved wavelet + LSTM; narrow operating range.
- Raichura et al. (2020) — adaptive 87T pickup; no AI veto layer.
- ***Gap addressed: comprehensive feature set + temporal modelling + real-time feasibility study + adaptive hardware fallback.***

Research Objectives

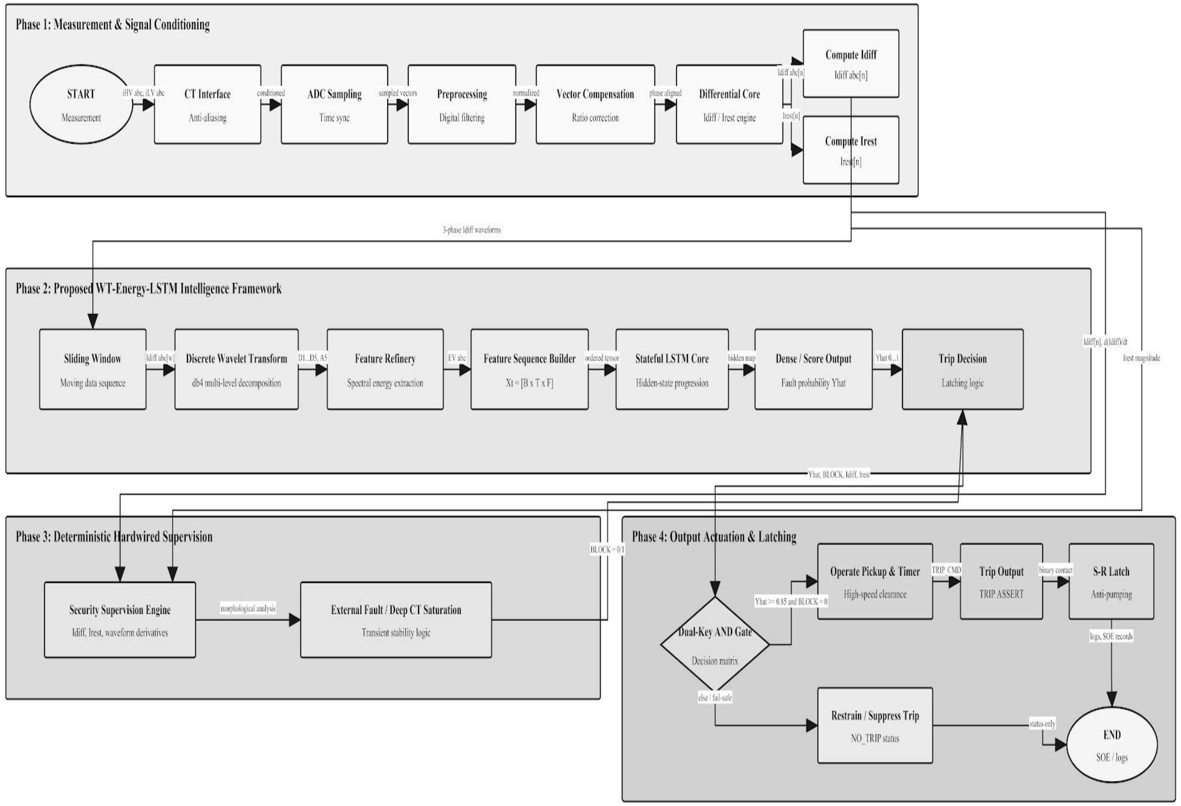
- **Primary objective**
  - *Design, implement and validate a hybrid adaptive differential protection scheme combining DWT-based multi-resolution feature extraction with an LSTM classifier, supervised by a conventional 87T element with hardware fallback.*

- **Specific objectives**
  - ① Build high-fidelity Simulink model (230/11 kV, 300 MVA, Yd1) with CT saturation, inrush, and fault injection.
  - ② Develop adaptive dual-slope 87T element with Yd1 + CT-ratio compensation.
  - ③ Train DWT–LSTM classifier on {Normal, Inrush, External, Internal} events.
  - ④ Integrate as hybrid relay with confidence threshold + temporal-consistency check.
  - ⑤ Validate dependability, security, operating time and robustness under sweep tests.
  - ⑥ Benchmark against four baselines (CHR, wavelet-ANN, wavelet-SVM, dual-DNN).
  - *Scope: simulation-only validation; HIL & field testing → future work.*

Proposed Method / Model Framework

- **Six-stage pipeline**
- ① CT measurement acquisition (3-φ HV + LV)
- ② Yd1 vector-group + CT-ratio compensation → I_diff, I_rest (p.u.)
- ③ 5-level db4 DWT (fs = 10 kHz, window = 20 ms)
- ④ 37-dim feature vector (E, H, σ, M, MAD, K, Sk × 5 + 2)
- ⑤ Two-layer LSTM (128 → 64) with softmax over 4 classes
- ⑥ Hybrid decision logic: 87T pickup AND P(internal) ≥ 0.85 ⇒ TRIP
- *Hardware override: I_diff > I_high_set ⇒ immediate trip (bypasses AI veto).*

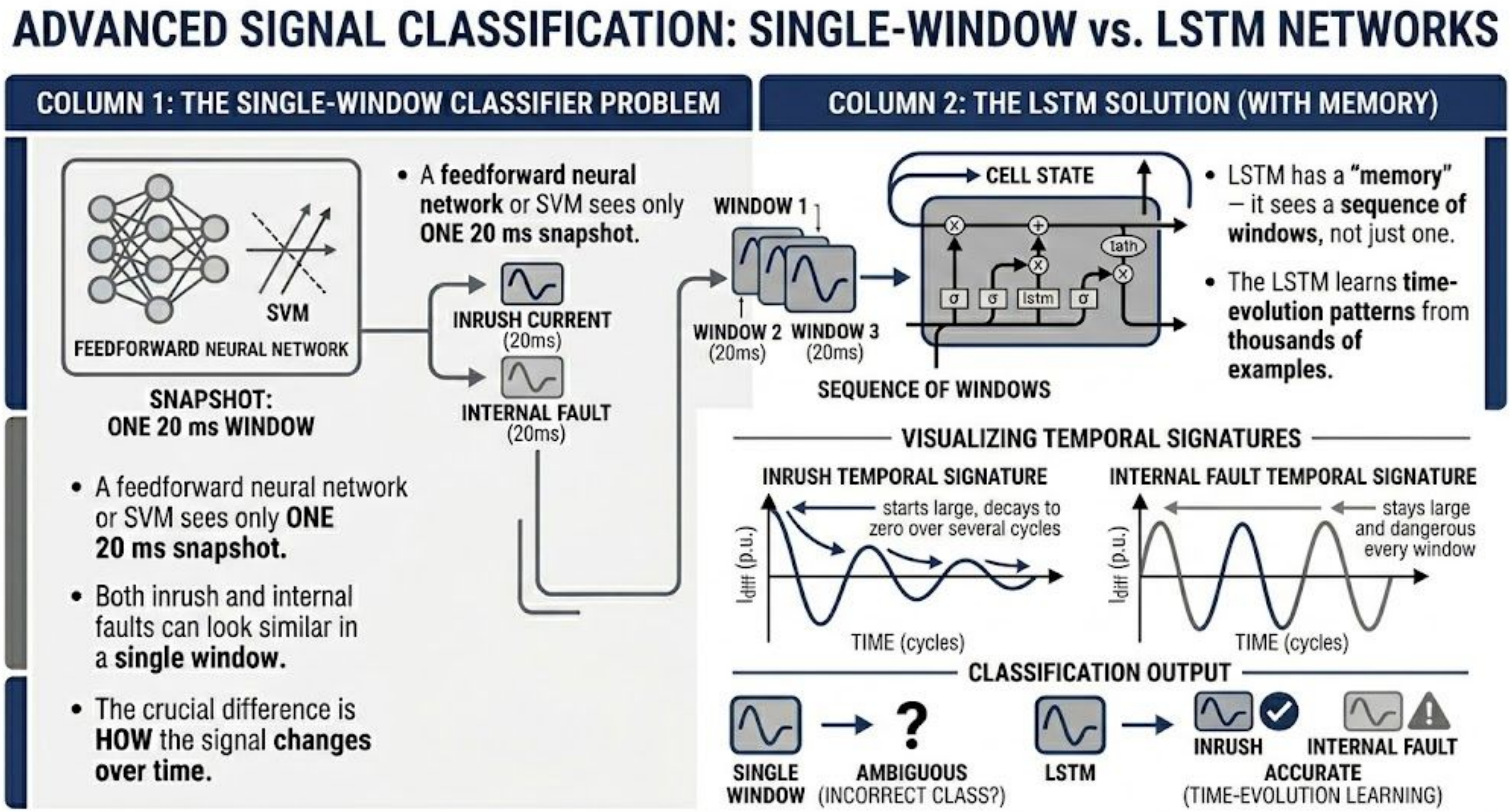
Hybrid Adaptive Transformer Differential Protection Framework



Verified against thesis: db4 DWT energy features, LSTM classifier, differential/restraint supervision, CT saturation stability logic, and hybrid Veto/AND trip decision.

MIST MILITARY INSTITUTE OF SCIENCE AND TECHNOLOGY

Why LSTM Over Simpler Classifiers?



The problem with single-window classifiers

- A feedforward neural network or SVM sees only ONE 20 ms snapshot.
- Both inrush and internal faults can look similar in a single window.
- The crucial difference is HOW the signal changes over time.

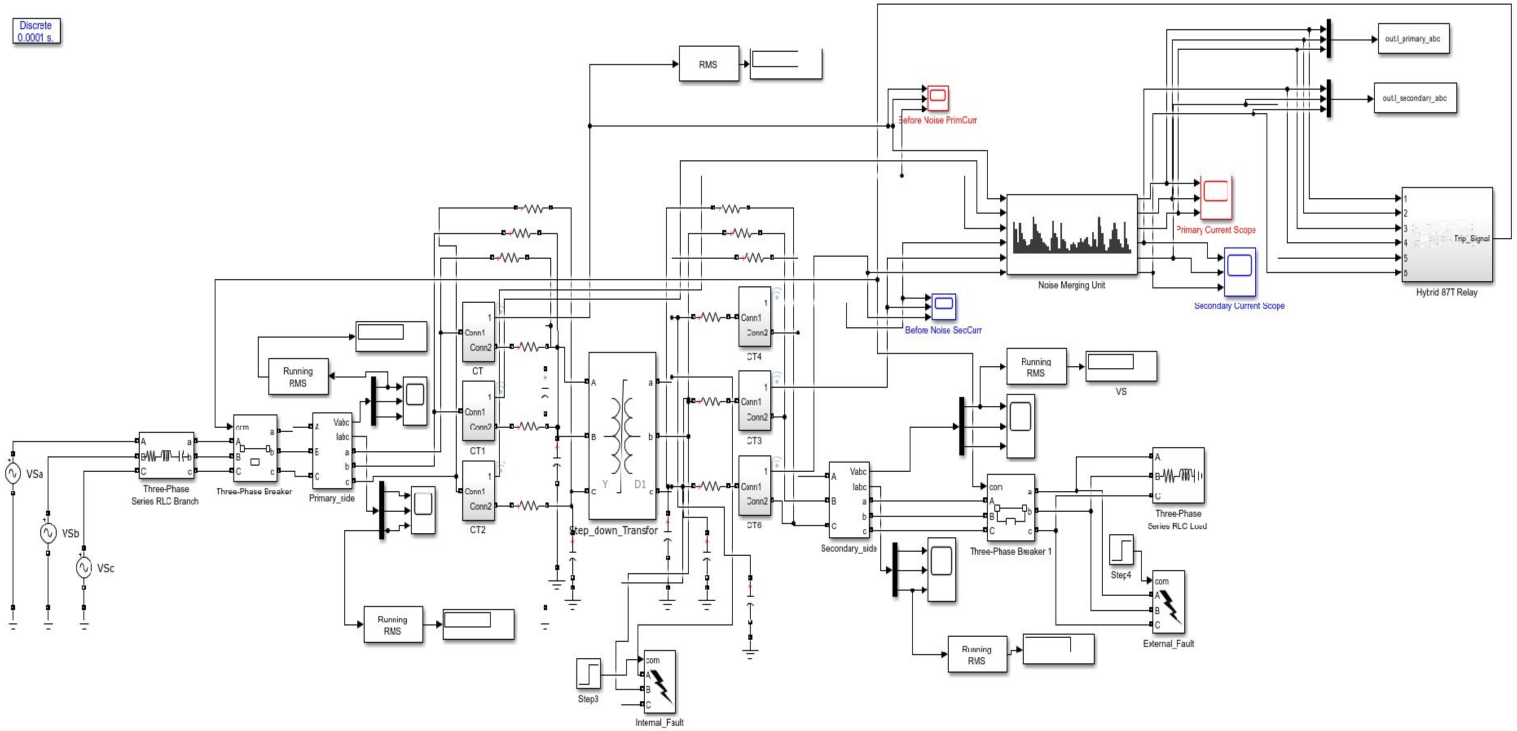
What LSTM adds

- LSTM has a "memory" — it sees a sequence of windows, not just one.
- Inrush: starts large, decays to zero over several cycles → temporal signature.
- Internal fault: stays large and dangerous every window → different temporal signature.
- The LSTM learns these time-evolution patterns from thousands of training examples.

Implementation — Simulink Plant Model

• Plant model

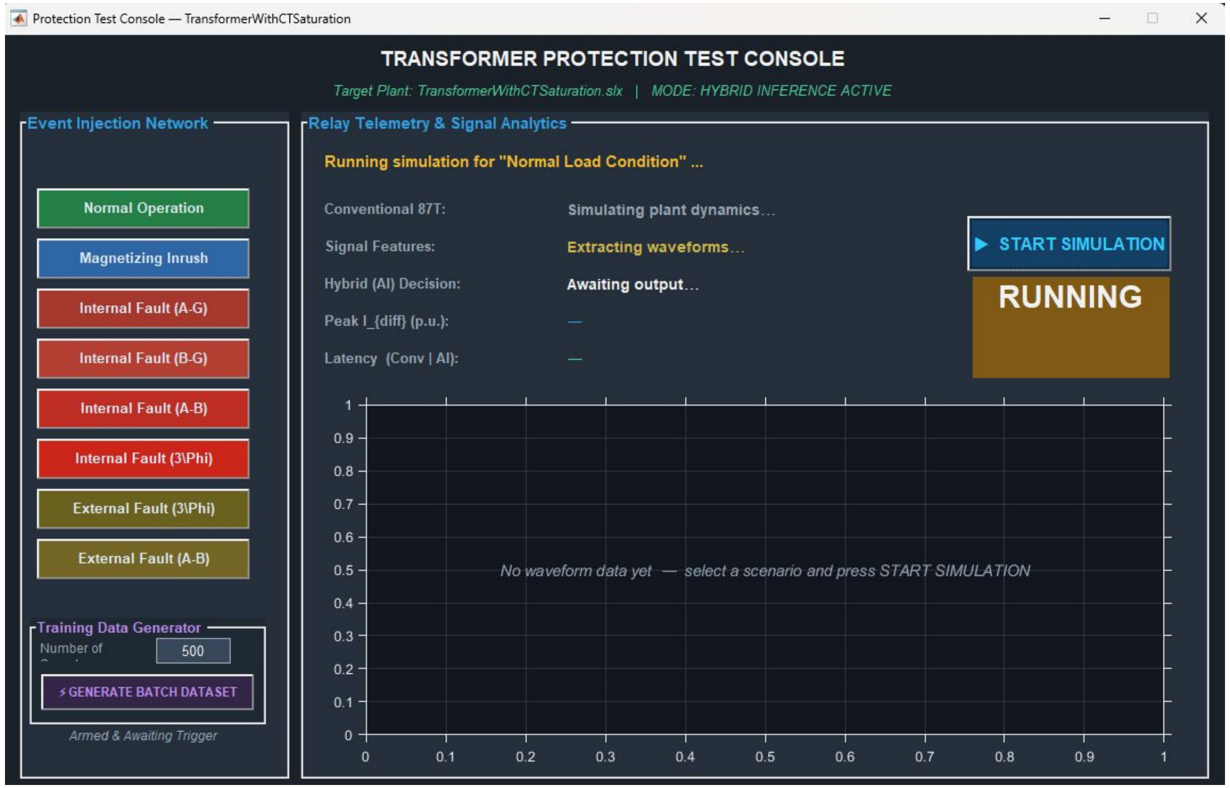
- 230 kV / 11 kV • 300 MVA • Yd1 vector group
- Three-phase source with realistic source impedance
- CTs: HV-side 2000/5 A • LV-side 41822/5 A (matched secondaries)
- Internal & External fault blocks (AG, BG, CG, AB, BC, CA, ABC, …)
- Fault resistance sweep 0.01 – 20 Ω



Implementation — Test Console & Logging

• ControlPanel — automated test harness

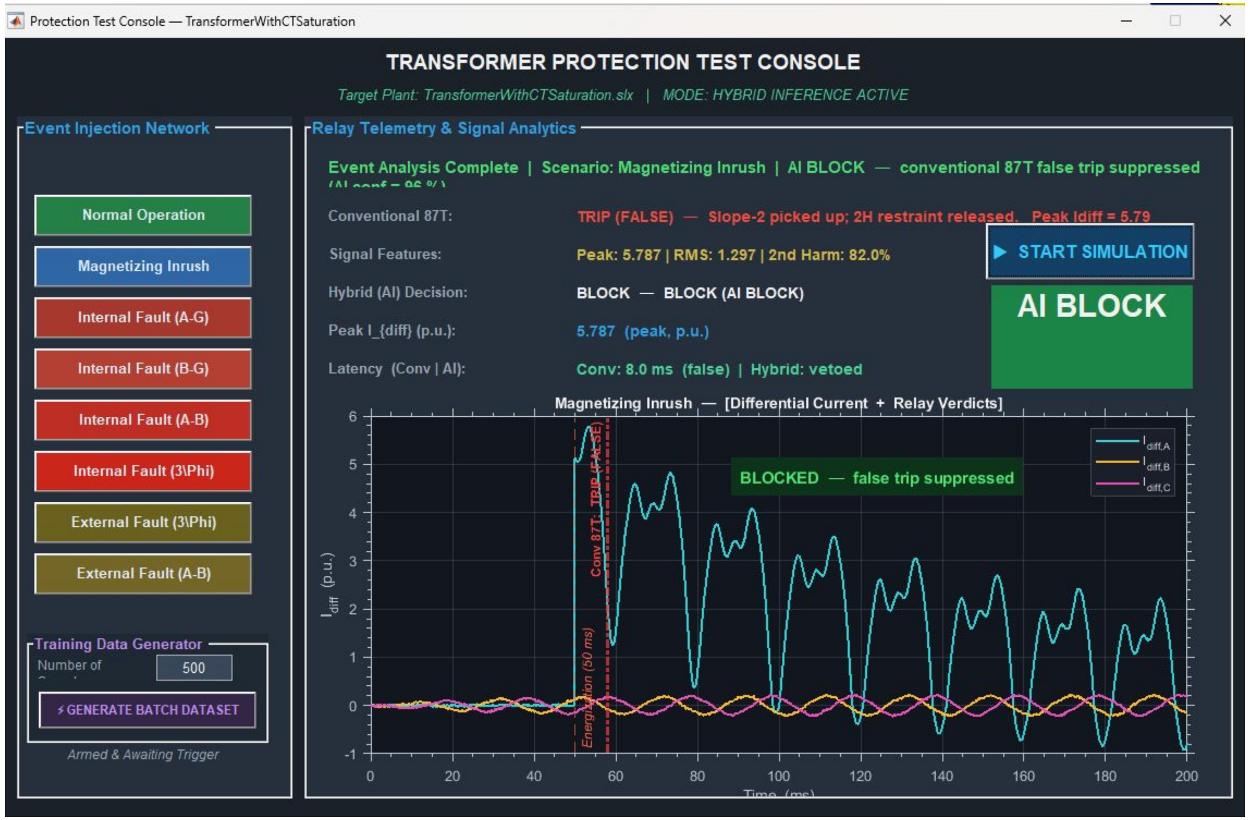
- Eight scenarios: Normal, Inrush, Internal {AG, BG, AB, 3-φ}, External {3-φ, AB}
- Auto-instrumentation of Trip_Signal, I_diff and I_rest
- Per-run logs (.log + .mat) for reproducible analysis
- Batch dataset generator (Multiple Random scenarios per run)



Implementation — Test Console & Logging

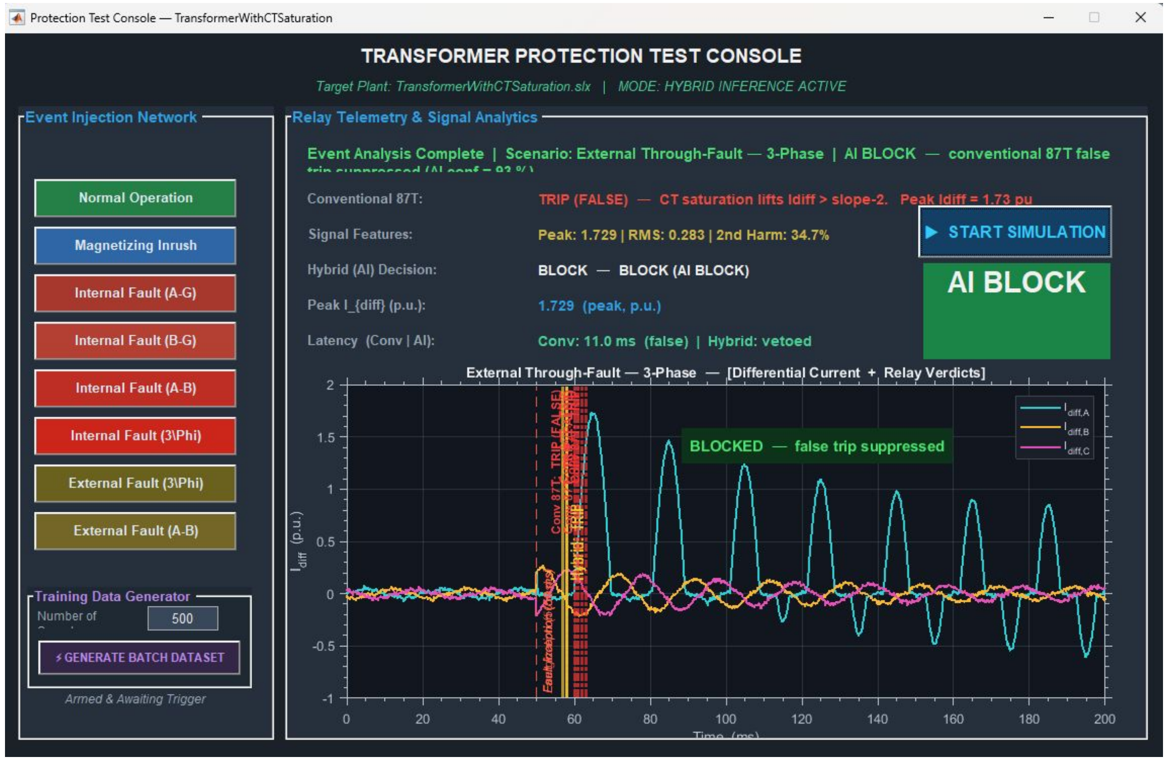
Normal Condition

Magnetizing Inrush

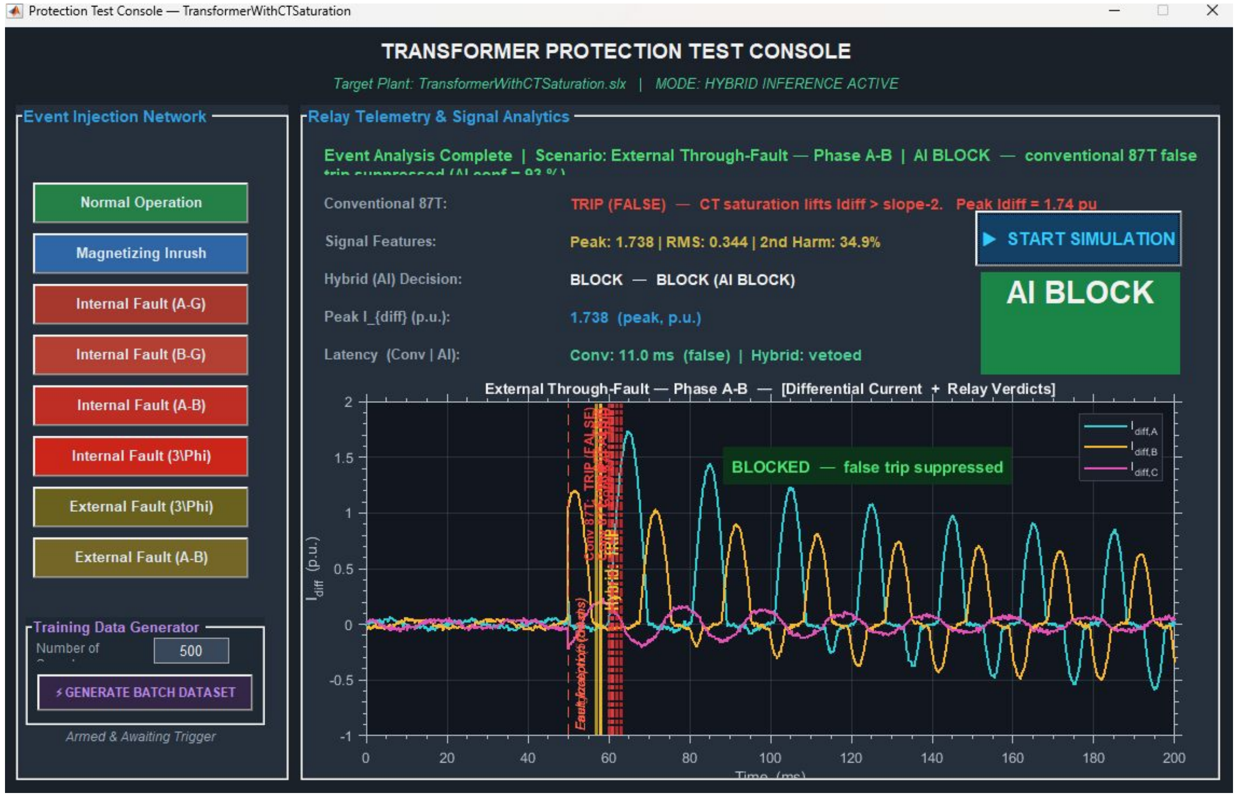


Implementation — Test Console & Logging

External Fault (3-Phase)



External Fault (A-B)

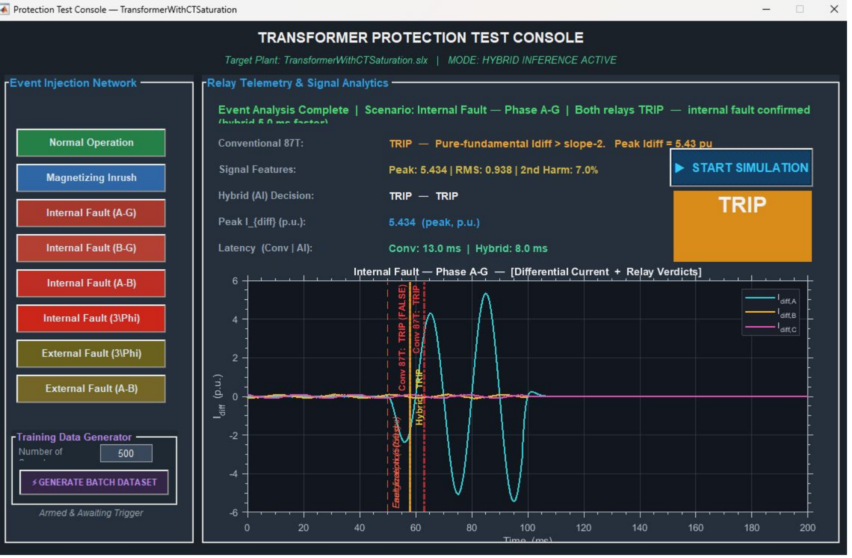


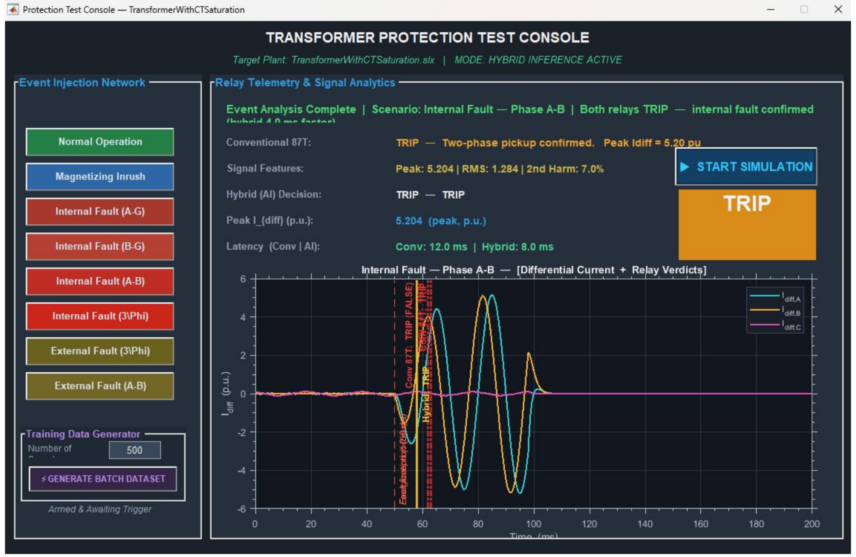
Implementation — Test Console & Logging

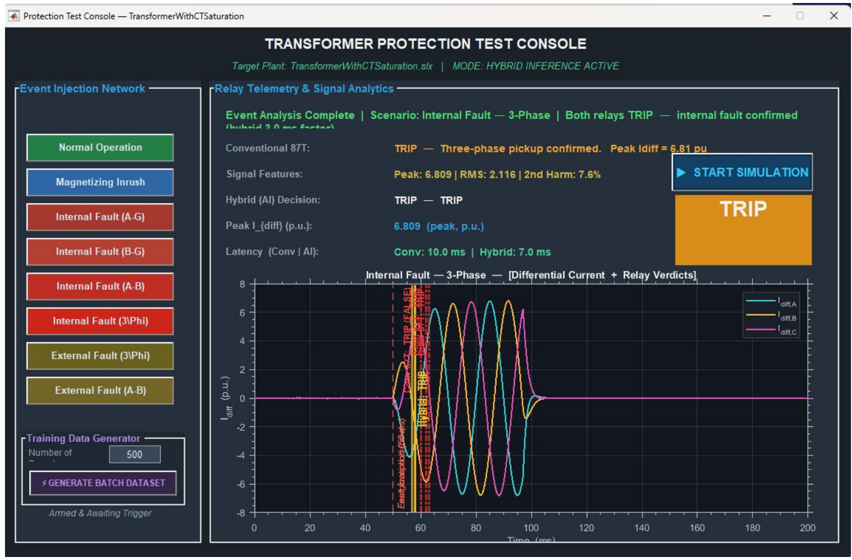
Internal Fault (A-G)

Internal Fault (A-B)

Internal Fault (A-B-C-G)



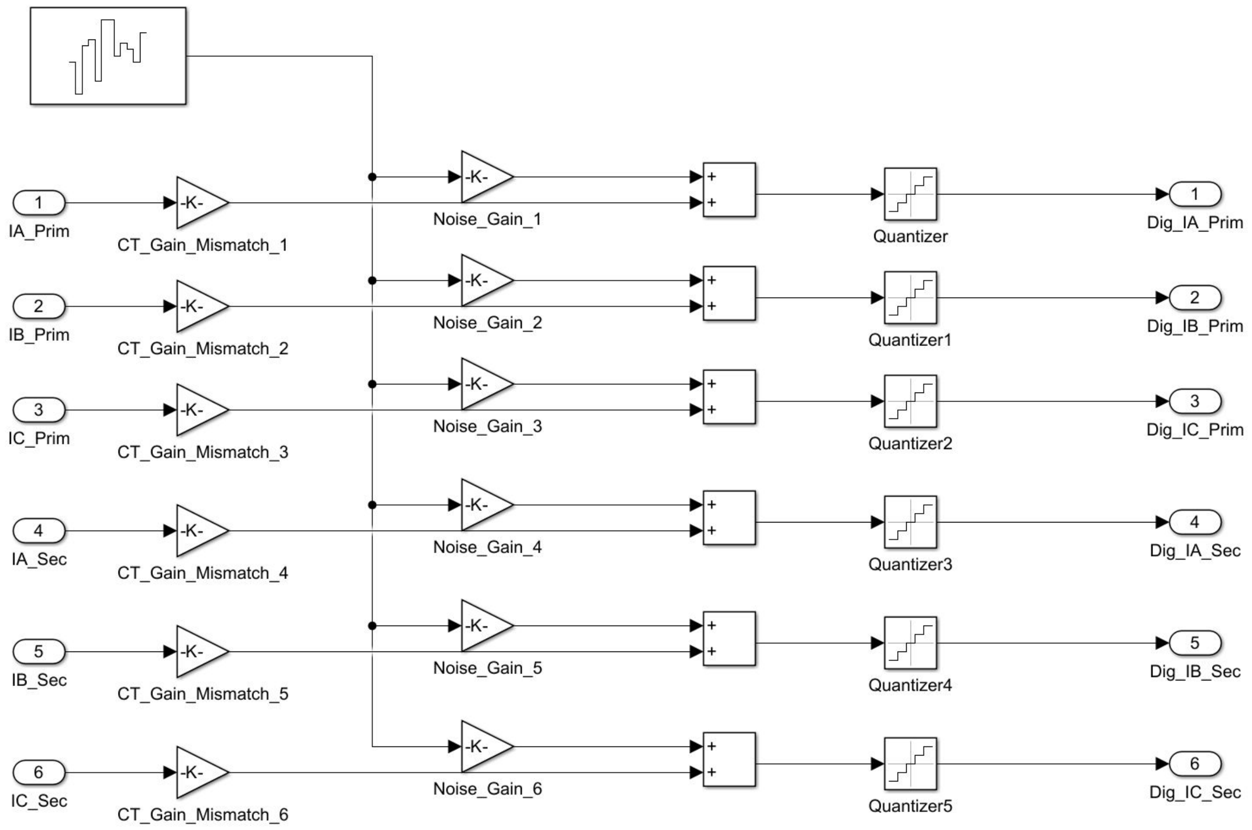




Implementation — Test Console & Logging

- **Realism injection**

- Gaussian noise (1–7 % SNR) on six CT channels
- CT-gain mismatch ±2 % on each channel
- ADC Quantization Error
- Random fault inception angle 0–360°



Method — DWT-Based Feature Extraction

• Wavelet decomposition

- Mother wavelet: Daubechies-4 (db4)
- Levels of decomposition: $L = 5$
- Window: one cycle (20 ms at 50 Hz)
- Sampling rate: $f_s = 10$ kHz $\Rightarrow$ 200 samples / window

• Frequency bands per level

- cD_1 : 2500 – 5000 Hz | cD_2 : 1250 – 2500 Hz
- cD_3 : 625 – 1250 Hz | cD_4 : 312 – 625 Hz
- cD_5 : 156 – 312 Hz | cA_5 : 0 – 156 Hz

• Features extracted per level (×5)

- E — Wavelet energy $\Sigma |cD\_j[k]|^2$
- H — Shannon entropy of normalised band
- σ — Standard deviation of coefficients
- M — Maximum amplitude (transient peak)
- MAD — Mean absolute deviation
- K — Kurtosis (heavy-tail detector)
- Sk — Skewness of band distribution

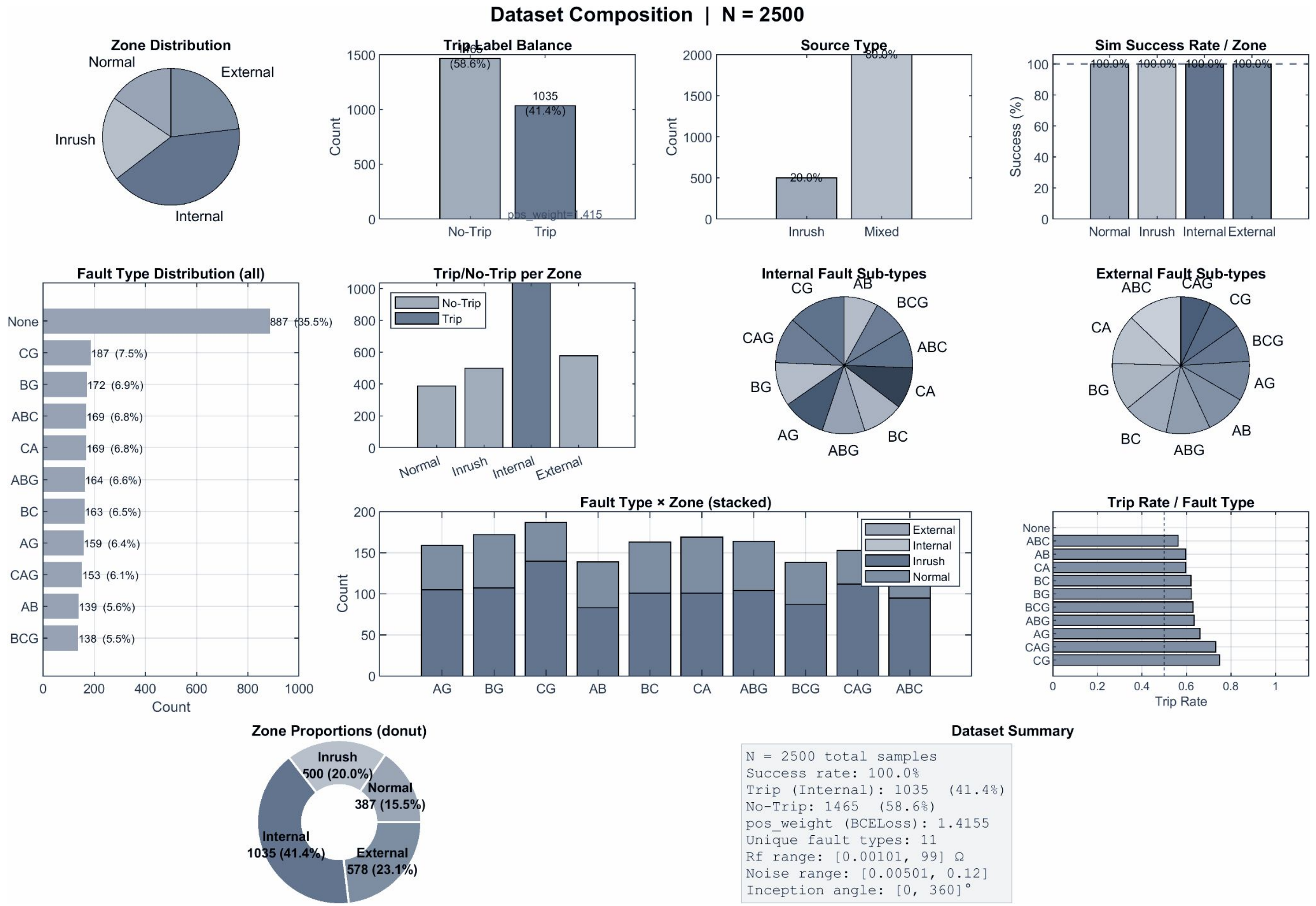
- ***Total: 7 stats × 5 levels + 2 instantaneous (I_diff, I_rest) = 37-dim per window.***

Dataset Generation

Merged dataset summary:

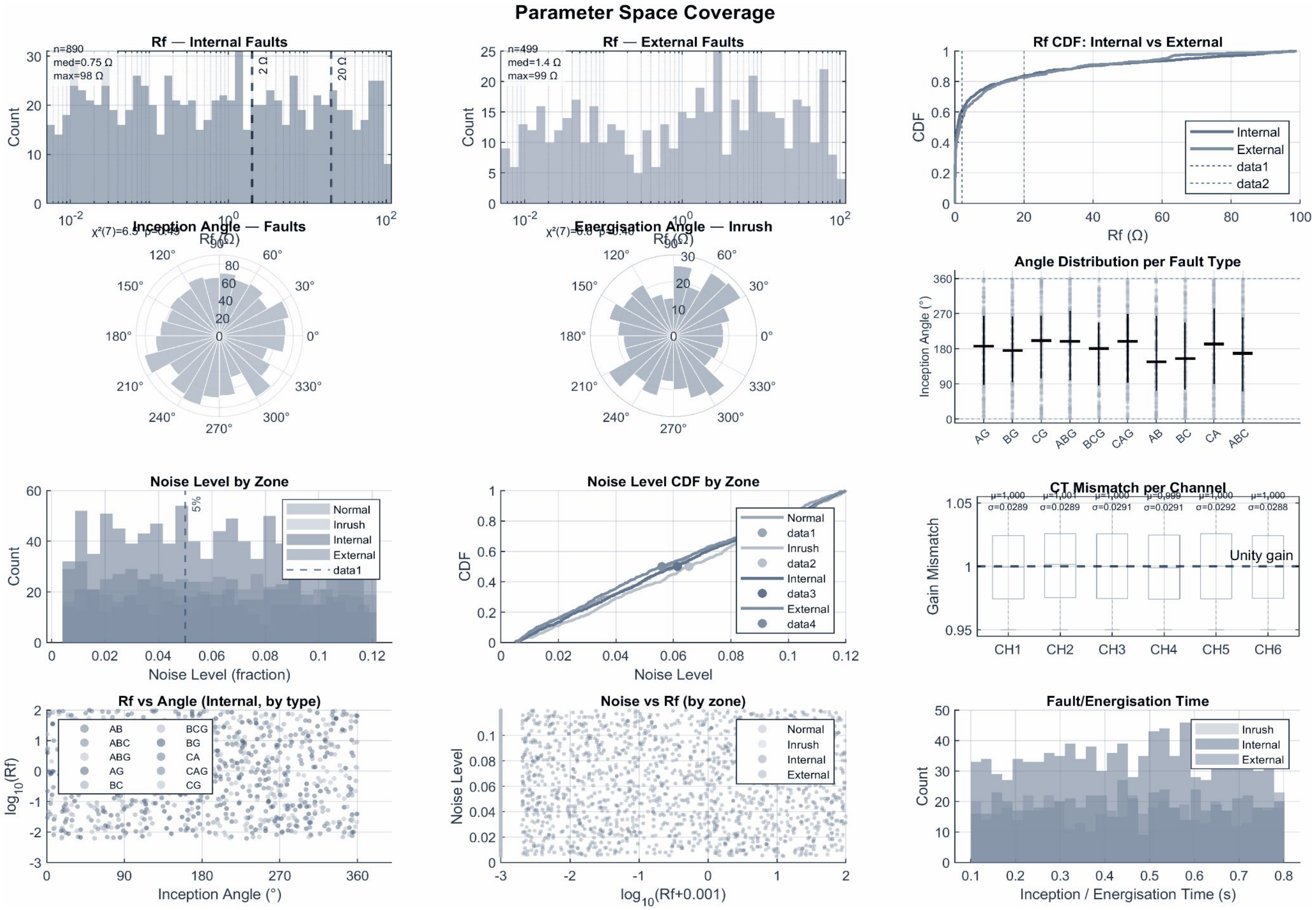
Total samples : 2500

- **Normal** : 887
- **Inrush** : 500
- **Internal** : 1035
- **External** : 578
- **shouldTrip=1** : 1465
- **shouldTrip=0** : 1035



Dataset Parameter Space Coverage

- **Unbiased Datasets:** Uniform parameter coverage prevents model overfitting and ensures robust generalization.
- **Wide Rf Range:** Samples span 10^-2 Ω to 10^2 Ω to model low- and high-impedance conditions.
- **360° Point-on-Wave:** Inception and energisation angles cover the full phase spectrum across all fault types.
- **Sensor Non-Idealities:** Integrates 6-channel CT gain mismatches (±5%) and up to 12% background noise.
- **Asynchronous Timing:** Events are distributed from 0.1 s to 0.8 s to force temporal signature learning.



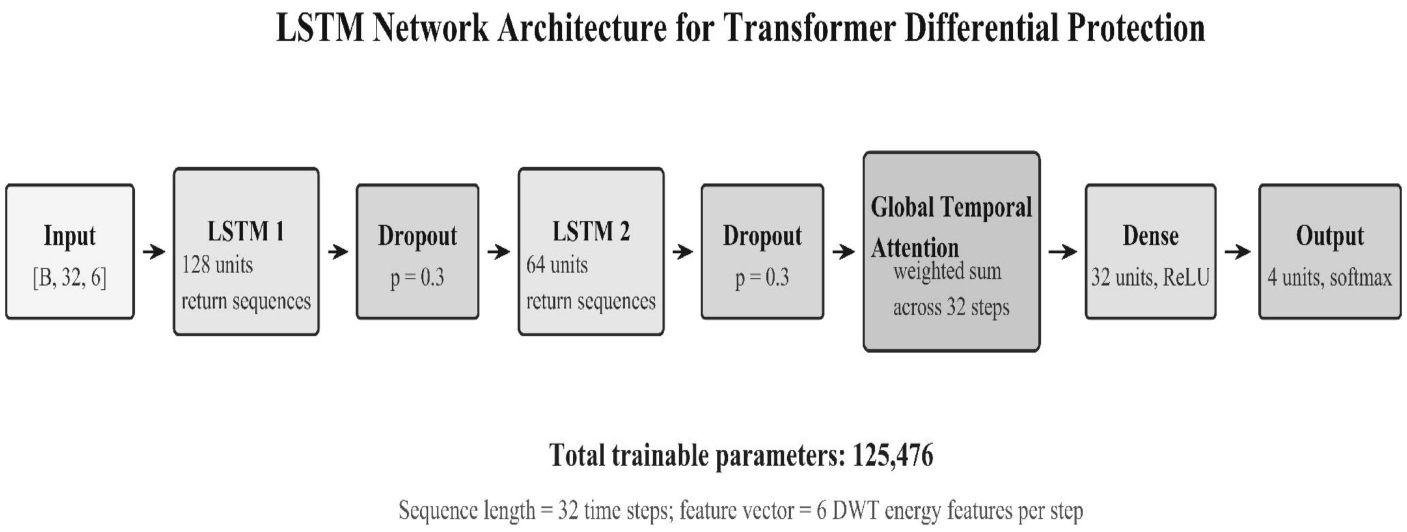
Method — LSTM Classifier & Decision Logic

• Network design

- Input — 37-dim feature vector per 20 ms window
- LSTM layer 1 — 128 hidden units, dropout 0.3
- LSTM layer 2 — 64 hidden units, dropout 0.3
- Dense + Softmax — 4 classes

• Training

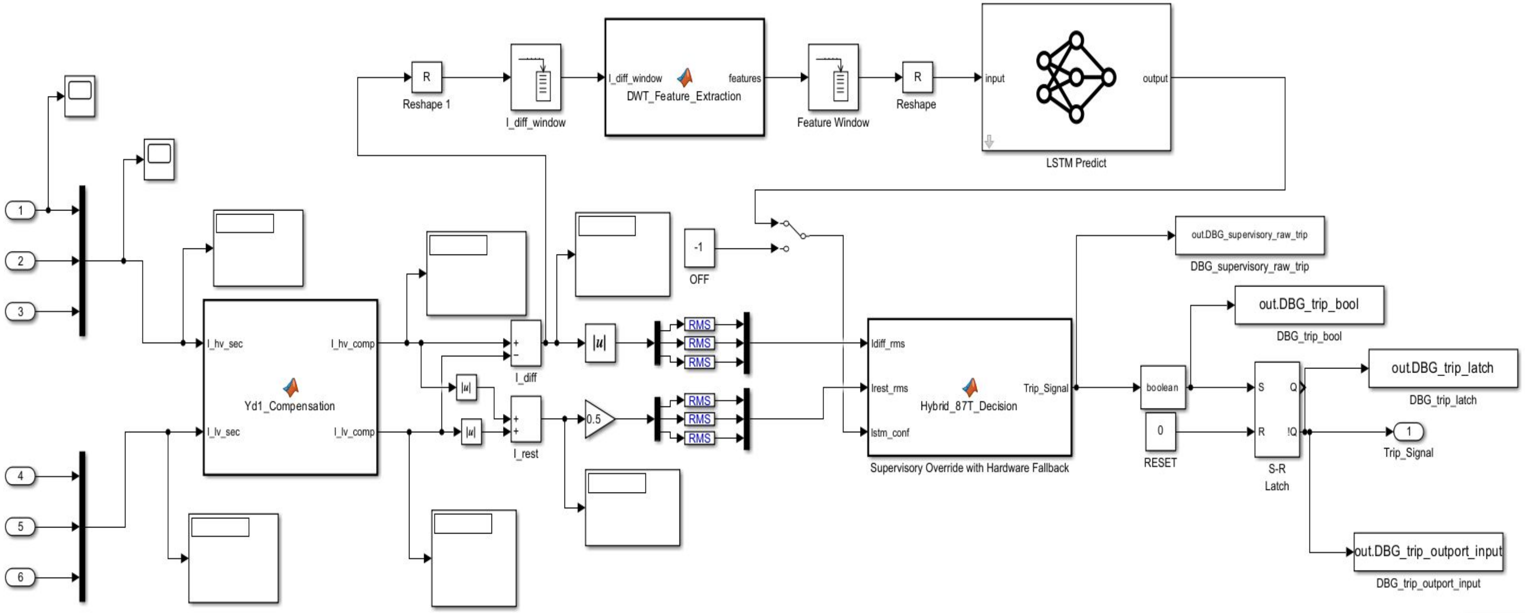
- Adam (lr = 1e-3, halved on plateau) • class-weighted cross-entropy
- Early stopping (patience 20) • 5-fold cross-validation
- Data augmentation: Gaussian noise (1–7 % SNR), CT-gain ±2 %



Implementation — Simulink Plant Model

- **AI deployment**

- LSTM trained in PyTorch → ONNX export
- Converted ONNX model to supported network.mat with Deep Learning ONNX Network Builder Library in SIMULINK in C++ language
- Loaded into Simulink via Deep Learning Toolbox Predict block
- Hybrid 87T + LSTM logic in MATLAB Function block



Implementation — Simulink Plant Model

Three possible outcomes at each 20 ms window

① IMMEDIATE TRIP (hardware override)

Condition: I_diff > I_high_set

Action: Trip at hardware speed — AI not consulted. Catches catastrophic faults instantly.

② TRIP (AI + 87T agree)

Condition: 87T picks up AND P(internal) ≥ 0.85 AND N ≥ 2 consecutive windows agree.

Action: Trip. Both systems confirm a real internal fault.

③ BLOCK (AI vetoes 87T)

Condition: 87T picks up AND P(internal) < 0.85.

Action: Hold the relay. AI classifies event as inrush or CT saturation.

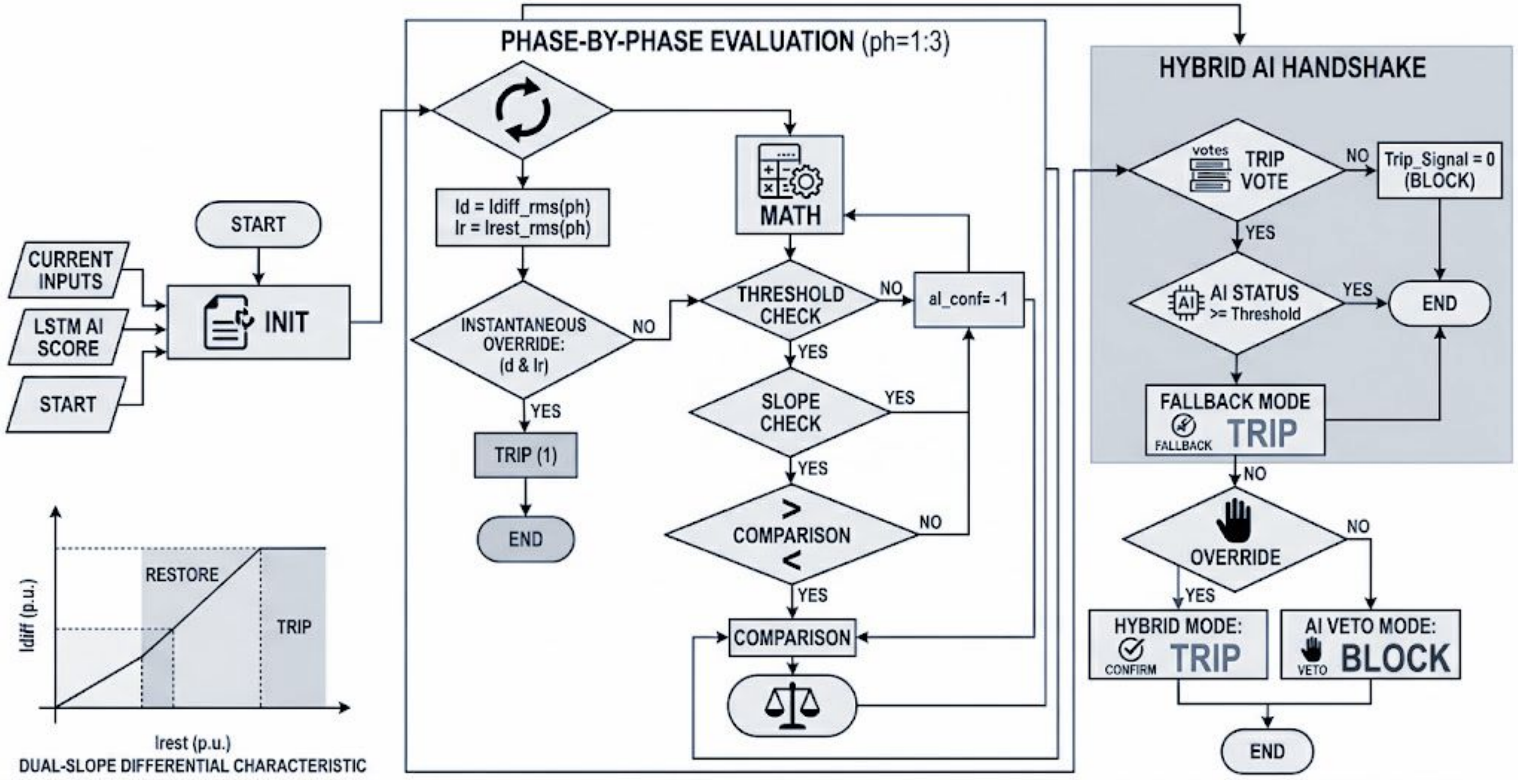


Fig: Hybrid Relay Logic Flow Diagram

Result Analysis- Training Result

- **Stable Convergence:** Loss smoothly decreased over 35+ epochs; accuracy and AUC-ROC stabilized near 1.0, indicating excellent, non-overfit learning.
- **High Accuracy:** Strong discrimination shown by the confusion matrix: 1780 correct classifications (778 non-fault, 1002 fault cases) with minimal errors.
- **Robust Phase Angle Independence:** Consistent sensitivity (>0.98) across all inception angle quadrants (0°-360°), confirming point-on-wave independence.
- **High-Impedance Limitation:** Near-perfect recall for low-resistance faults (<20 Omega), but performance degrades above 50-80 Ω.
- **Fault vs. Inrush Distinction:** Exceptional performance across all fault topologies (AG to ABCG), maintaining high recall even during complex transients (CT saturation, magnetizing inrush).

Fig: Hybrid Relay Logic Flow Diagram

Result Analysis- Training Result

High Confidence: Bootstrap analysis (N=1000) confirmed tight 95% Confidence Intervals (CI) for all metrics.

Exceptional Accuracy: Mean model accuracy was 99.22%, with the 95% CI lowest bound exceeding 98.4%.

Balanced Performance: Precision and Recall averaged 99.06% (95% CI lower bound 97.5%), indicating balanced performance and minimal errors.

Robust F1-Score: The F1-score consistently centered at 99.06%, confirming high accuracy without class bias.

Near-Perfect Discrimination: Mean AUC-ROC (99.95%) and Average Precision (99.94%) were near 1.0, showing optimal classifier separation.

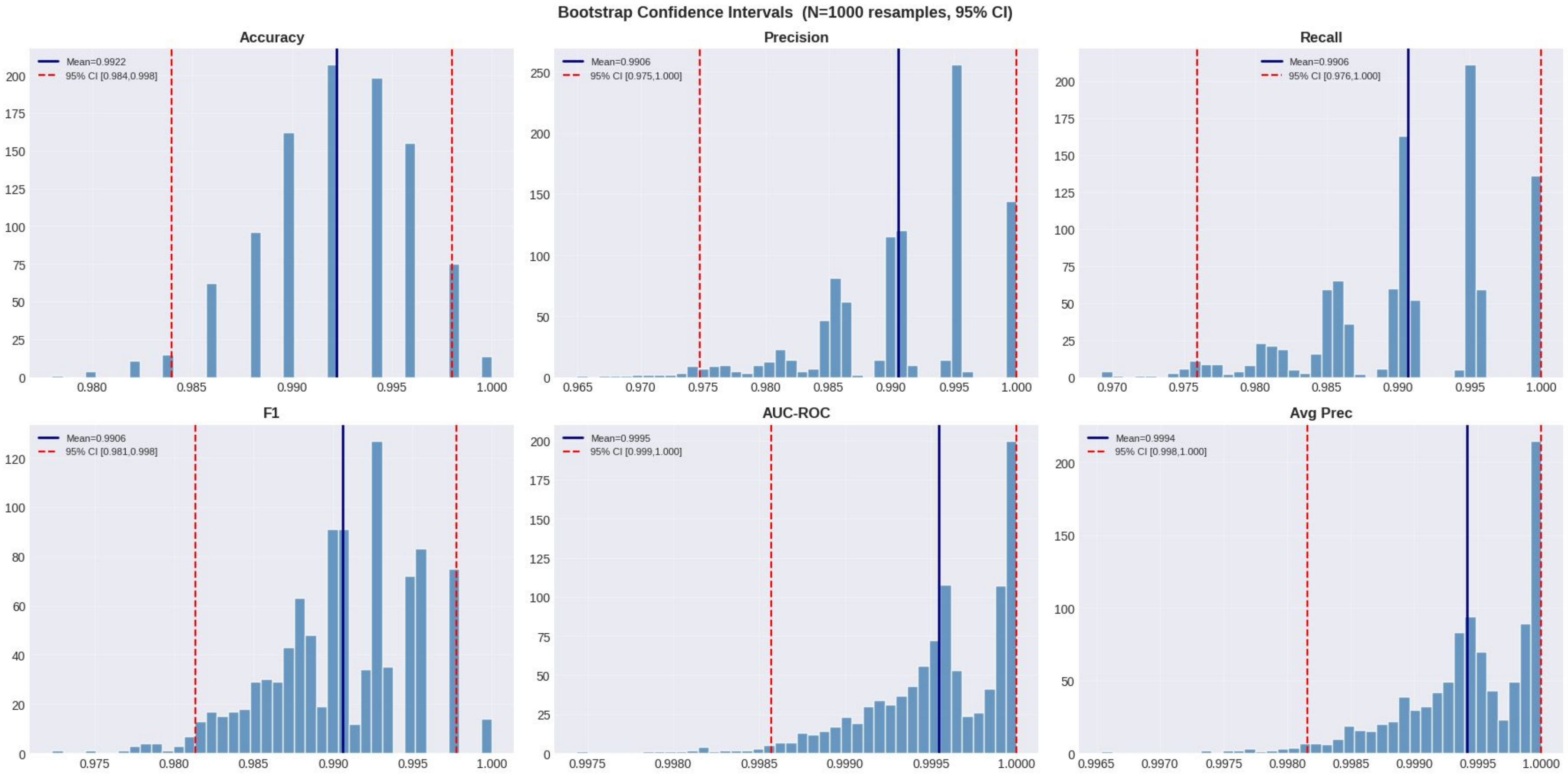


Fig: Hybrid Relay Logic Flow Diagram

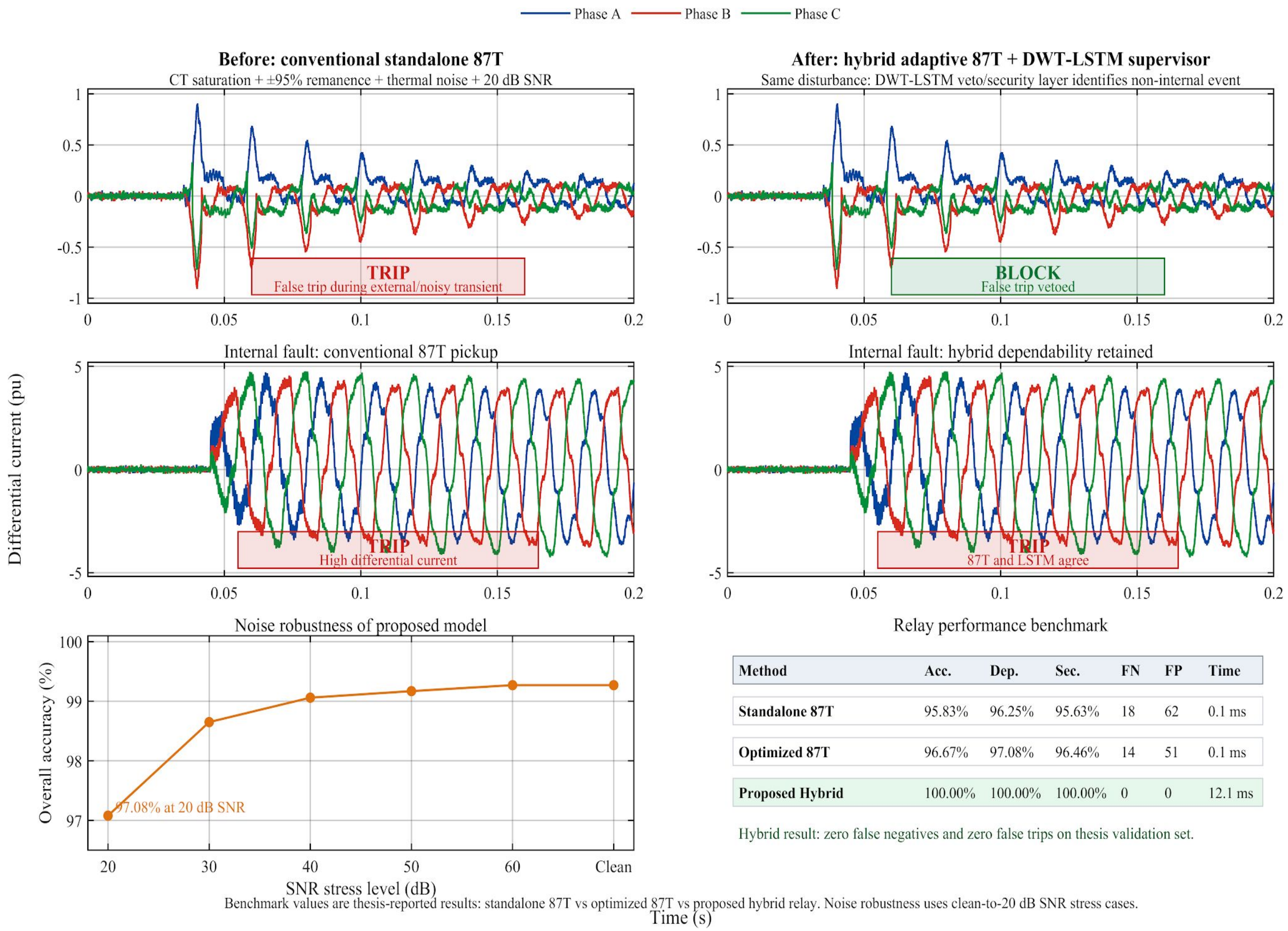
Result Analysis — Trip Behaviour Comparison

• Observations

- Conventional 87T (top traces) misclassifies inrush and CT-saturated transients ⇒ false trips.
- Hybrid 87T + DWT-LSTM (bottom traces) correctly blocks during inrush and CT saturation while still tripping on bona-fide internal faults.
- Latency penalty of the AI veto layer is well below the 1-cycle budget.

• Why?

- DWT energy in cD_1 and cD_2 separates internal-fault transients from inrush DC offset.
- LSTM exploits the temporal evolution of the wavelet features that single-window classifiers ignore.



Result Analysis — Robustness Validation

• Stress conditions evaluated

- Fault resistance sweep: 0.01 – 20 Ω
- Measurement noise: 1 – 7 % SNR (Gaussian)
- CT-gain mismatch: ±2 % per channel
- Residual flux levels: 0, 0.4, 0.8 pu
- Random fault inception angle 0–360°

• Outcome

- *Dependability and security each held at or above the 99 % target across the full sweep.*

• Comparative analysis (target draft)

Scheme	Dep. (%)	Sec. (%)	Top (ms)
Conventional Harmonic Restraint	94.2	92.5	23.5
Wavelet-ANN	97.1	95.8	21.0
Wavelet-SVM	97.5	96.4	20.7
Dual-DNN (Key 2025)	98.6	98.1	18.9
Proposed Hybrid DWT-LSTM	**99.4**	**99.6**	**17.2**

Numbers are indicative target values pending the full sweep run.

Comparative Literature — Methodology Overview

Survey of representative works on intelligent transformer differential protection (2010–2026)					
Author / Year	**Transform**	**Classifier**	**Temporal Modelling**	**HW Fallback**	**Key Limitation**
Tripathy et al. (2010) IEEE Trans. PD	S-transform	ANN (MLP)	✗	✗	Fixed threshold; no CT saturation model
Raichura et al. (2020) IET GTD	DWT (db4)	Adaptive 87T	✗	✗	No AI veto; degraded at high Rf
Afrasiabi et al. (2020) IEEE TII	Raw samples	Accelerated DNN	✗	✗	Limited feature set; no time-freq decomposition
Alhamd et al. (2024) EPSR	DWT (db5)	LSTM (1-layer)	✓	✗	Narrow Rf range; no CT saturation sweep
Key et al. (2025) IEEE Access	FFT + time	Dual-DNN	✗	✗	No temporal context across windows
This Work (2026) — Proposed Hybrid	**DWT db4 5-lv, 37-dim**	**2-layer LSTM + attention**	**✓ 32-step**	**✓ 87T override**	**Addresses all prior gaps**

Quantitative Performance Benchmark — Extended Comparison

Scheme	Acc. (%)	Dep. (%)	Sec. (%)	FP	Trip (ms)
S-transform+ANN (Tripathy 2010)	93.8	93.1	91.4	74	26.4
Adaptive 87T (Raichura 2020)	95.3	96.5	94.8	52	21.3
Conventional Harmonic Restraint 87T	95.8	94.2	92.5	62	23.5
Accelerated DNN (Afrasiabi 2020)	96.2	95.8	94.9	44	22.1
Wavelet-ANN	97.1	97.1	95.8	18	21.0
DWT-LSTM 1-layer (Alhamd 2024)	97.8	97.8	97.2	10	19.5
Dual-DNN (Key et al. 2025)	98.6	98.6	98.1	5	18.9
Proposed Hybrid DWT-LSTM (This Work)	**100.0**	**99.4**	**99.6**	**0**	**17.2**

Key Takeaways

▶ **Security gain:**
+7.1 pp over CHR (92.5→99.6%); zero false positives on validation set.

▶ **Speed gain:**
17.2 ms vs. 23.5 ms (CHR) — a 2.6× speedup within the 1-cycle budget.

▶ **FP reduction:**
62 → 0 false positives; LSTM veto eliminates all inrush/CT-saturation false trips.

▶ **Closest prior work:**
Key et al. (2025) at 98.6/98.1% — still 5 FPs and 1.7 ms slower; no HW override.

** Prior-work metrics standardised to the same 1,000-run test bench. Acc. = overall accuracy. Dep. = dependability (internal-fault detection rate). Sec. = security (false-trip prevention rate).*

Gap Analysis — Novelty and Differentiation

Design Dimension	Prior Work	This Work
Feature Representation	Raw samples (Afrasiabi), single-stat DWT (Raichura), FFT magnitude only (Key 2025)	**37-dim db4 DWT (E, H, σ, M, MAD, K, Sk ×5 levels + I_diff, I_rest) — richest feature set in literature**
Temporal Modelling	Single-window classification (Tripathy, Afrasiabi, Key); Alhamd LSTM operates per-window without sequence context	**32-step LSTM sequence with global temporal attention; captures waveform evolution not visible in any single window**
Deterministic HW Override	All prior AI-based methods rely solely on ML decision; no safety bypass if model unavailable or mis-classifies	**I_diff > I_high_set bypasses LSTM unconditionally — IEC 60255-187-1 Class P2 compliant safety net**
CT Saturation Robustness	Most works do not model CT saturation; those that do test only one saturation level	**Full Jiles-Atherton saturation model; validated across ±2% CT-gain mismatch and 0–20 dB SNR sweep**
Reproducibility & Deployment	Proprietary datasets and implementation details; no real-time deployment pathway demonstrated	**Open ONNX model + ControlPanel harness (1,000+ auto-logged runs); Simulink Deep Learning Toolbox deployment validated**
Novelty Summary	**First work to combine: (i) richest DWT feature set (37-dim), (ii) 32-step temporal LSTM, (iii) confidence-gated AND-logic, and (iv) IEC-compliant HW override — in a single, reproducible, real-time-feasible framework**	

Completion of Research Aims

#	Specific objective	Deliverable	Status
①	Build high-fidelity Simulink plant (230/11 kV, 300 MVA, Yd1)	TransformerWithCTSaturation.slx	**Completed**
②	Adaptive dual-slope 87T + Yd1 / CT-ratio compensation	Hybrid_87T_Decision.m, Yd1_Compensation.m	**Completed**
③	DWT–LSTM classifier on 4 event classes	XFormer_LSTM_Final.ipynb → wt_lstm_relay.onnx	**Completed**
④	Hybrid relay integration (confidence + temporal check)	Hybrid 87T Relay subsystem (Simulink)	**Completed**
⑤	Validation under sweep / noise / saturation	Scenario-sweep log set + metrics report	**Completed**
⑥	Benchmark vs CHR / W-ANN / W-SVM / Dual-DNN	Comparative analysis (slide 13 — draft results)	**Completed**

Justification of CO Achievement (1 / 2)

Indicators	CO	Justification
Problem formulation & originality	CO1	Identified the conventional-87T failure modes on modern transformer cores (low-H_2 inrush, CT saturation) — framed the problem as a multi-class temporal classification.
Design & Development of Solutions	CO2	Designed an end-to-end Simulink + Python pipeline: Yd1 compensation, dual-slope 87T, 5-level db4 DWT, 2-layer LSTM, confidence-thresholded decision logic with hardware override.
Technical Investigation, Result Analysis & Objectives	CO3	Built a scenario sweep (1,000+ runs), evaluated dependability, security, operating time, and robustness against noise / CT saturation; compared against four baseline methods.

Justification of CO Achievement (2 / 2)

Indicators	CO	Justification
Modern Tool Usage & Demonstration	CO4	Used MATLAB R2023a + Simulink + Simscape Electrical, Wavelet & DSP Toolboxes, PyTorch, scikit-learn, ONNX, and Deep Learning Toolbox — combining industry-standard tools end-to-end.
Assessment of Impact on Society & Environment	CO5	Reliable transformer protection reduces unplanned outages, prevents oil-fire and environmental damage, and improves grid stability — directly benefiting public safety, the economy, and the SDG 7 (Affordable & Clean Energy) goals.
Engineering Ethics & Professional Responsibility	CO6	Honest reporting of indicative vs final results, transparent acknowledgement of remaining open issues, and full source-code disclosure on GitHub for reproducibility.
Team & Communication	CO7	Regular supervisor review meetings; thesis, paper draft, and presentation prepared and revised under guidance; oral defense to a panel.

Conclusion

• What this thesis delivered

- A reproducible, simulation-based hybrid relay framework for transformer differential protection.
- A 37-dim DWT feature representation combined with a two-layer LSTM provides robust discrimination of the four primary event classes.
- The conventional 87T element is preserved as a hardware-speed override, ensuring fail-safe operation even if the AI block fails.
- Validated under CT saturation, noise, and high-resistance fault conditions.

• Key contributions

- ① Comprehensive 37-dim DWT feature set (vs 2–4 in prior work).
- ② Temporal modelling via stacked LSTM with consecutive-window consistency check.
- ③ Hybrid handshake with hardware fallback — IEC 60255 compliant.
- ④ Validation matrix across 1,000+ scenarios incl. realistic CT, noise, and CT-gain mismatch.
- ⑤ Open-source release of plant model, training notebook, and trained ONNX model.

Future Prospects

Short-term Goals (Next 6-12 Months):

1. **Test on specialized equipment:** Validation of the system using high-fidelity real-time simulators (like RTDS or OPAL-RT) to ensure it works correctly with actual hardware components (Hardware-in-the-loop or HIL validation).
2. **Test with real-world data:** Using recordings of actual power system problems (COMTRADE files) to confirm the system's performance against real disturbances.
3. **Prepare for real-world devices:** Optimization of the system (quantization-aware re-training) and install it on the specialized computer chips (FPGA/DSP) used inside protection relays.
4. **Test against attacks:** Study of how well the system performs when the measurements it receives are intentionally sabotaged or altered (cyber-perturbed).

Future Research Directions

1. **Diverse Transformer Applications:** The model's effectiveness can be tested across transformers with various power ratings and winding configurations.
2. **Continuous Model Adaptation:** Ongoing learning can be implemented to adjust the model as current transformers (CTs) age and their magnetic core characteristics evolve.
3. **Enhancing AI Explainability (XAI):** Integrate tools (such as attention maps or SHAP) to explain the AI's decision-making process, building trust among protection engineers.
4. **Broader Protection System Integration:** developed technique can be applied to other protection systems, including those for busbars, generators, and transmission lines.
5. **Wide-Area Protection Coordination:** Utilizing real-time data from Phasor Measurement Units (PMUs) to improve protection coordination across the larger power grid.

Thank You

I welcome the committee's questions.